Structure Factor report

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In this report are the main results of the structure factor analysis.

Results Experiment 1

An exponential fit, $f(|\vec{q}|) = \frac{A}{|\vec{q}|^{\alpha}}$, is performed to the structure factor of the systems shown in the following figure **??**.

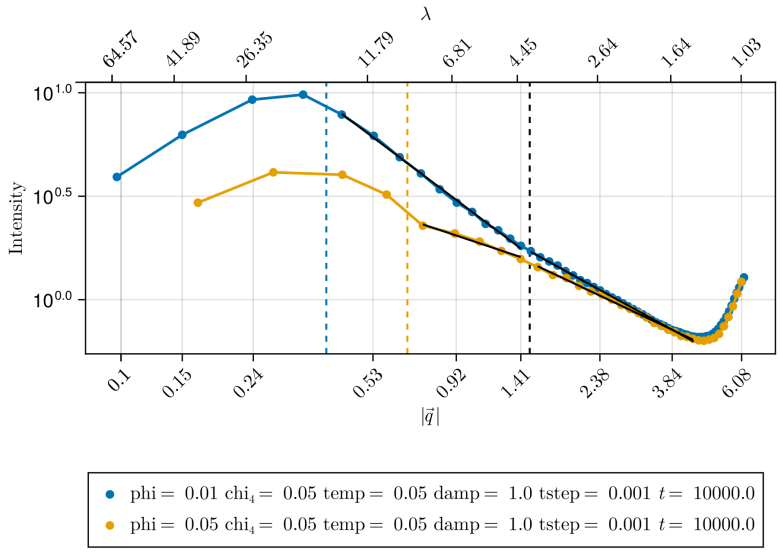


Figure 1: Structure factor of two systems with different packing fraction at the same time instant.

The result of the parameters are shown in the following table **??**. The systems 1 and 2 are represented with blue and yellow colours repectively. Finally, zone 1 is defined as $\lambda_1 \in [7/2\sigma_p, 21/2\sigma_p]$ and zone two is defined as $\lambda_2 \in [21/2\sigma_p, L/4]$.

System	Zone	A	α
1	1	0.925179	2.486543
2	1	0.794425	2.08354
1	2	1.264535	2.731100
2	2	0.561453	1.948924

Table 1: Results of the fit.

Discussion

For me it is not clear the interpretation of α. On the other hand, it is intereset to see that the structure factor can be analyzed in three domains.

Results Experiment 2

An exponential fit, $f(|\vec{q}|) = \frac{A}{|\vec{q}|^{\alpha}}$, is performed to the structure factor of the systems shown in the following figure **??**.

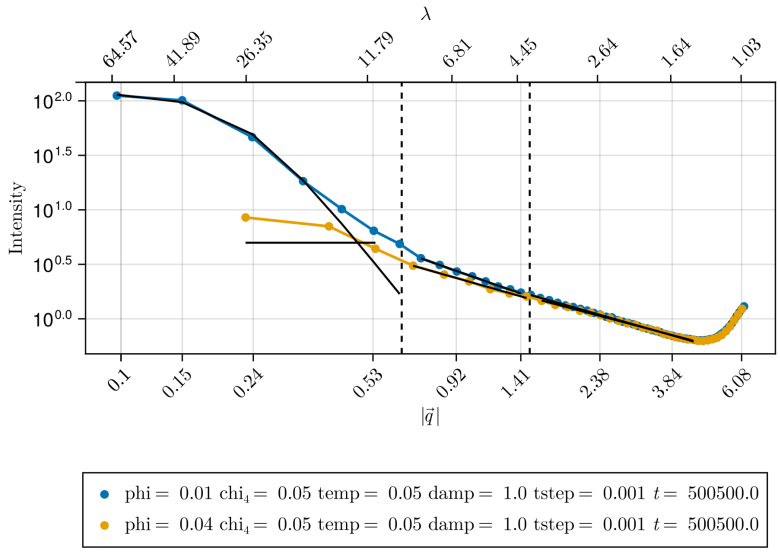


Figure 2: Structure factor of two systems with different packing fraction at the same time instant.

The result of the parameters are shown in the following table **??**.

System	Zone	A	α
1	1	0.925179	2.486543
2	1	0.794425	2.08354
1	2	1.264535	2.731100
2	2	0.561453	1.948924

Table 2: Results of the fit.

[0.910700366460469, 2.400446307859152] [1.1219562891705535, 2.51040488815733693] [118.43465362627254, 4.000113568664359, 4.557270807249773]

[0.8454990176654736, 2.2140825372770747] [0.9070017450863235, 2.196358167662408] [4.992773078000656, 37.44053418672465, 0.40000000000034347]

Discussion

For me it is not clear the interpretation of α. On the other hand, it is intereset to see that the structure factor can be analyzed in three domains.

Characteristic distances in the system and the domains for the fit

The distance range for the pair-wise interaction is defined by the distance at minimum energy, to the cutoff, $r_b \in [\sigma_p, 3/2\sigma_p]$. Then, considering that the center of all particles, central and patch, lies on the same line the distances between the center of the central particles is

given by $r_{\text{int}} = 2\sigma_p + r_b$. If we evaluate r_b at the limits, the interaction distance is $r_{\text{int}} \in [1.2, 1.4]$. Finally, if we increase the amount of patchy particles in pair-wise interaction, the distance between central particles is given by

$$d_{\text{int}} = (N-1)\left(2\sigma_p + r_b\right).$$

Zone definition of Experiment 1

The result fo this analysis is used to defined the region at small distances. More specifically, the lower limit of this region is set at $N = 2$ and $r_b = 3/2\sigma_p$ and the upper limit is set to $N = 4$ and $r_b = 3/2\sigma_p$. Leading to the following spatial domain,

$$\lambda_1 \in \left[7/2\sigma_p, 21/2\sigma_p\right].$$

Then, we define a second spatial domain, which it's upper limit is set the order of magnitud of the total length of the simulation box, as follows,

$$\lambda_2 \in \left[21/2\sigma_p, L/4\right].$$

Zone definition of Experiment 2